Integration by Substitution and Trignometric identities

(Standard u-substitution and Trigonometric Substitution)

Substitution method

Integration by substitution is the reverse of the chain rule. If $u = g(x)$ is a differentiable function, then

$$\int f(g(x))g'(x)\,dx = \int f(u)\,du.$$

Trigonometric substitution is a specific case used for integrals involving expressions of the form $\sqrt{a^2 - x^2}$, $\sqrt{x^2 - a^2}$, or $\sqrt{x^2 + a^2}$. The idea is to replace x with a trigonometric function to simplify the radical.

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Problems

I. Standard u-Substitution

S1. Compute

$$\int (2x + 1)^5\,dx$$

using $u = 2x + 1$.

S2. Evaluate

$$\int \frac{\ln x}{x}\,dx$$

with $u = \ln x$.

S3. Compute the definite integral

$$\int_0^1 xe^{x^2}\,dx$$

by substituting $u = x^2$.

S4. Solve

$$\int \frac{dx}{\sqrt{1 + 5x}}$$

by setting $u = 1 + 5x$.

II. Trigonometric Substitution

T5. Compute

$$\int \frac{dx}{\sqrt{9 - x^2}}$$

using the substitution $x = 3\sin\theta$.

Why $x = 3\sin\theta$ works for $\sqrt{9-x^2}$

When we encounter the integral

$$\int \frac{dx}{\sqrt{9-x^2}},$$

we notice the square root $\sqrt{9-x^2}$ requires that the expression under the radical is non-negative:

$$9 - x^2 \geq 0 \quad \Longrightarrow \quad -3 \leq x \leq 3.$$

To simplify the square root, we use a trigonometric substitution that satisfies

$$1 - \sin^2\theta \geq 0.$$

Specifically, we set

$$x = 3\sin\theta \quad \Longrightarrow \quad dx = 3\cos\theta\, d\theta.$$

Then the radical becomes

$$\sqrt{9-x^2} = \sqrt{9 - 9\sin^2\theta} = \sqrt{9(1-\sin^2\theta)} = 3\cos\theta.$$

This substitution works nicely because the identity $1 - \sin^2\theta = \cos^2\theta$ ensures the expression under the square root is always non-negative for $\theta \in [-\pi/2, \pi/2]$, which corresponds to the original range $x \in [-3, 3]$.

T6. Evaluate

$$\int \frac{dx}{x^2\sqrt{x^2-4}}$$

with $x = 2\sec\theta$.

T7. Compute

$$\int \sqrt{x^2+16}\, dx$$

using $x = 4\tan\theta$.

T8. Evaluate the definite integral

$$\int_0^1 \frac{dx}{\sqrt{1-x^2}}$$

using the substitution $x = \sin\theta$.

T9. (Challenge) Compute

$$\int \frac{dx}{x\sqrt{x^2-1}}$$

Why Trigonometric Substitutions Work

Trigonometric substitutions simplify integrals involving radicals of the forms $\sqrt{a^2 - x^2}$, $\sqrt{x^2 + a^2}$, and $\sqrt{x^2 - a^2}$. The key idea is to choose a substitution that automatically satisfies the non-negativity of the expression under the square root.

1. **For $\sqrt{a^2 - x^2}$:** Set $x = a\sin\theta$, $dx = a\cos\theta\, d\theta$.
   - Requirement: $a^2 - x^2 \geq 0 \implies -a \leq x \leq a$.
   - Substitution: $x = a\sin\theta \implies -1 \leq \sin\theta \leq 1$, automatically satisfying the domain.
   - Radical simplifies: $\sqrt{a^2 - x^2} = \sqrt{a^2 - a^2\sin^2\theta} = a\cos\theta$.
2. **For $\sqrt{x^2 + a^2}$:** Set $x = a\tan\theta$, $dx = a\sec^2\theta\, d\theta$.
   - Requirement: $x^2 + a^2 \geq 0$ — always true, but substitution helps simplify the radical.
   - Substitution: $x = a\tan\theta \implies dx = a\sec^2\theta\, d\theta$.
   - Radical simplifies: $\sqrt{x^2 + a^2} = \sqrt{a^2\tan^2\theta + a^2} = a\sqrt{\tan^2\theta + 1} = a\sec\theta$.
3. **For $\sqrt{x^2 - a^2}$:** Set $x = a\sec\theta$, $dx = a\sec\theta\tan\theta\, d\theta$.
   - Requirement: $x^2 - a^2 \geq 0 \implies |x| \geq a$.
   - Substitution: $x = a\sec\theta \implies \sec\theta \geq 1$, automatically satisfying the domain.
   - Radical simplifies: $\sqrt{x^2 - a^2} = \sqrt{a^2\sec^2\theta - a^2} = a\sqrt{\sec^2\theta - 1} = a\tan\theta$.

Summary: The substitutions are chosen so that the Pythagorean identities

$$1 - \sin^2\theta = \cos^2\theta, \quad 1 + \tan^2\theta = \sec^2\theta, \quad \sec^2\theta - 1 = \tan^2\theta$$

allow the radicals to simplify nicely, while the domains of the trigonometric functions automatically satisfy the conditions imposed by the square roots.

Hints and Tips

- Always identify the inner function (or radical expression) that suggests substitution.
- For trigonometric substitution, recall the Pythagorean identities:

$$1 - \sin^2\theta = \cos^2\theta, \quad \sec^2\theta - 1 = \tan^2\theta, \quad 1 + \tan^2\theta = \sec^2\theta.$$

- Don't forget to convert back to x after integrating in θ.
- For definite integrals, you can either change the limits to θ or revert to x before evaluating.

Key Trigonometric Identities and a Memory Trick

Pythagorean Identities:

$$\begin{aligned} \sin^2\theta + \cos^2\theta &= 1, \\ 1 + \tan^2\theta &= \sec^2\theta, \\ \cot^2\theta + 1 &= \csc^2\theta. \end{aligned}$$

Angle-Sum Identities:

$$\begin{aligned}\sin(A+B) &= \sin A\cos B + \cos A\sin B,\\ \cos(A+B) &= \cos A\cos B - \sin A\sin B,\\ \tan(A+B) &= \frac{\tan A+\tan B}{1-\tan A\tan B}.\end{aligned}$$

Double-Angle and Other Useful Identities:

$$\begin{aligned}\sin(2\theta) &= 2\sin\theta\cos\theta,\\ \cos(2\theta) &= \cos^2\theta-\sin^2\theta = 2\cos^2\theta-1 = 1-2\sin^2\theta,\\ \tan(2\theta) &= \frac{2\tan\theta}{1-\tan^2\theta}.\end{aligned}$$

Euler's Formula Memory Trick: Recall Euler's formula:

$$e^{i\theta} = \cos\theta + i\sin\theta, \quad \text{so also } e^{-i\theta} = \cos\theta - i\sin\theta.$$

So, $e^{i(A+B)} = \cos(A+B) + i\sin(A+B) = e^{iA}e^{iB} = (\cos A + i\sin A)(\cos B + i\sin B)$
How this helps:

- Squaring $\sin\theta$ and $\cos\theta$ gives the Pythagorean identity naturally: $(\cos\theta)^2 + (\sin\theta)^2 = 1$.
- Expressions like $\tan\theta = \frac{\sin\theta}{\cos\theta}$ and $\sec\theta = \frac{1}{\cos\theta}$ follow immediately.
- Angle-sum and double-angle formulas can be derived by expanding $(\cos\theta + i\sin\theta)^2 = e^{2i\theta}$ or $(\cos A + i\sin A)(\cos B + i\sin B) = e^{i(A+B)}$.

Summary: Using $e^{i\theta}$ allows you to derive and remember all Pythagorean, angle-sum, and double-angle identities algebraically rather than by rote memorization.

Derivatives of Basic Trigonometric Functions

Standard Trigonometric Derivatives:

$$\begin{aligned}\frac{d}{dx}[\sin x] &= \cos x,\\ \frac{d}{dx}[\cos x] &= -\sin x,\\ \frac{d}{dx}[\tan x] &= \sec^2 x,\\ \frac{d}{dx}[\cot x] &= -\csc^2 x,\\ \frac{d}{dx}[\sec x] &= \sec x\tan x,\\ \frac{d}{dx}[\csc x] &= -\csc x\cot x.\end{aligned}$$

C1. Compute $\int \sin^3 x\cos^2 x dx$
Hint: sin **is odd and** cos **is even**

C2. Compute $\int \tan^5 x \sec^4 x dx$

Think pair share if the method of integration will change if we replace $\tan^5 x$ with $\tan^8 x$ or any other power of tan?

C3. Compute $\int \tan^5 x \sec^7 x dx$

Method will not change for any power of $\sec x$ **as long as it is present, but power of** $\tan x$ **has to be odd. Why?**

C4. Compute $\int \tan^3 x \, dx$
Hint: Choose $u = \tan x$ **then** $du = \sec^2 x \, dx \implies dx = \frac{du}{1+u^2}$